



During soldering, lead fumes are generated, which are toxic. The flux used for soldering also emits fumes, which again are highly toxic if inhaled in excess. So it is always better to remove these toxic fumes from the working area.

Test point	Details
TP0	0V, GND
TP1	12V DC
TP2	High when soldering iron is in the stand

Semiconductors:	
IC1	- CA3130 operational amplifier
T2	- BC547 npn transistor
T2	- BD139 npn transistor
LED1	- 5mm red LED
D1	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 100-kilo-ohm
R2	- 100-ohm
R3, R4	- 1-kilo-ohm
VR1	- 10-kilo-ohm preset
Capacitors:	
C1	- 470µF, 35V electrolytic
Miscellaneous:	
CON1-CON2	- 2-pin male and female connector
PZ1	- Piezoelectric sensor
	- 12V, 7Ah battery
	- 12V small DC exhaust fan

The circuit described here removes these toxic fumes from the area automatically, when required, using a small exhaust fan.

Circuit and working

The circuit diagram of a solder fumes remover is shown in Fig. 1. The circuit is built around popular op-amp CA3130 (IC1), transistors BC547 and BD139 (T1 and T2, respectively), 12V

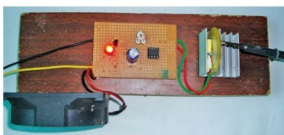


Fig. 1: Author's prototype

DC small exhaust fan and some easily-available components.

CA3130 is an excellent op-amp that requires very low input current. It is a 15MHz BiMOS IC with MOSFET inputs and a bipolar output. MOSFET transistors present in its inputs provide very high input impedance. The input current can be as low as 5 pA. Presence of these PMOS transistors results in a common-mode input voltage down to 0.5 volts below the negative rail. The internal bias of the IC at 5 pA is sufficient for its working, thus external biasing is not required in this case.

A heat sink with piezoelectric element PZ1

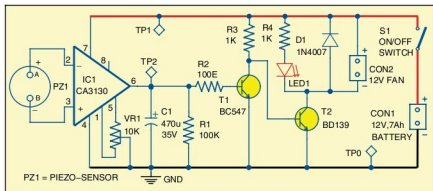


Fig. 2: Circuit diagram of solder fumes remover

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